

Internship Report

SRM University, AP

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Abstract

This internship focused on understanding Quantum Speed Limits (QSLs), which define the fundamental limits on the time required for a quantum system to evolve from one state to another. The project explored various formulations of QSLs, including the Mandelstam-Tamm bound and the Margolus-Levitin bound, and their applications in two state quantum systems.

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Chapter 1

Introduction

This chapter introduces the background of the internship, the motivation behind the project, and the main objectives that were set out to be achieved.

1.1 Background

Describe the context of your work at SRM AP.

1.2 Objectives

To learn about Quantum Speed Limits and apply it to a two state quantum system.

Chapter 2

Methodology

This chapter details the methods, tools, and techniques used during the internship.

2.1 Tools and Technologies

Mention the software, hardware, or theoretical frameworks utilized.

2.2 Implementation Details

Describe how the project was implemented step-by-step.

Chapter 3

Results and Conclusion

This chapter presents the results obtained from the experiments or tasks carried out during the internship.

3.1 Observations

Detail your findings here. You can include figures and tables.

3.2 Discussion

Discuss the significance of your results and how they meet the objectives.

3.3 Summary

Briefly restate the main achievements of the project.

3.4 Future Work

Mention any extensions or improvements that could be made to the project in the future.

Bibliography

- [1] John Doe and Jane Smith. An example paper on machine learning. *Journal of Examples*, 2026.